Specieslike clusters with several founders

Maximality, infinite species classes, and the complexity of the ancestor-point axiom

21 October 2026

Abstract

Alexander asks for constraints that put every organism in a maximal specieslike cluster without requiring an included common ancestor. We give two such choices: a founding-cohort condition and any fixed finite bound on the number of founders. Both retain chain closure and coverage; the second also permits founders born after non-founders. Their infinite maximal clusters represent exactly the same generator-equivalence classes as Alexander's common-ancestor clusters. The common proof identifies the role of finitely many founders in the maximality argument. We then examine what can be learned from finite genealogical observations. Global satisfaction of the identical ancestor point axiom is complete at the third Borel level, already for connected rooted trees of outdegree at most two. This descriptive complexity statement concerns a yes/no property of a biosphere. It does not assert impossibility of every form of species classification.

1. The coverage question

A genealogical definition of species faces two different tasks. It must say which sets of organisms qualify, and it must ensure that the definition actually covers the organisms under consideration. A collection of plausible local conditions need not give the second property: an organism can belong to successively larger qualifying sets without belonging to any maximal one.

Alexander's *Specieslike clusters based on identical ancestor points* [A] makes this issue precise. A specieslike cluster is a connected, ancestrally convex set satisfying the identical ancestor point axiom. That axiom prohibits a member from having both infinitely many descendants and infinitely many non-descendants inside the set. Alexander proves coverage after adding reflection and a common-ancestor condition. Following his Theorem 13, he asks for alternatives that do not require that common ancestor.

The common ancestor must itself belong to the cluster. This distinction matters for a population founded by several organisms: a cluster can have a unified subsequent history without containing one organism ancestral to all its founders. Our alternatives allow such populations.

The first condition requires every internal founder to be born before every internal non-founder. The second fixes a finite upper bound on the number of founders while placing no such restriction on their birthdates. Both guarantee coverage. Their difference is visible in the following genealogy:

$$a_0 \longrightarrow a_1 \longrightarrow a_2 \longrightarrow c_0 \longrightarrow c_1 \longrightarrow \cdots, \qquad b \longrightarrow c_0.$$

Take the birth order to be $a_0, a_1, a_2, b, c_0, c_1, \ldots$. The whole biosphere has two founders, a_0, b, and

no common ancestor. Founder b arrives after two non-founders. The fixed bound of two admits this biosphere; the founding-cohort condition excludes it.

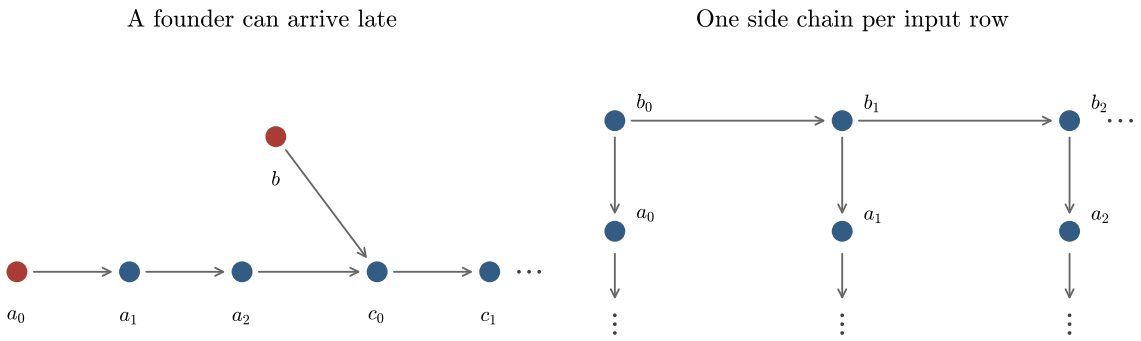


Figure 1: Two ways finite founding histories enter the argument. Left: a late founder joins an existing lineage. Right: the backbone and side chains used in the complexity proof. Arrows are parent-to-child.

Three results organize the paper. Section 4 proves coverage for both alternatives through one chain argument. Section 5 shows that they preserve the infinite species classes defined in [A]. Section 7 proves a descriptive-complexity theorem explaining a limit of finite observation, even when the genealogy has a single founder.

2. Biospheres, clusters, and founders

We use Alexander's infinite-biosphere assumptions [A, Definition 1]. The directed graph G has a real birthdate $t(v)$ for each vertex. Parents are born strictly before children; only finitely many vertices are born before any fixed real time; every vertex has finitely many children; and G is infinite. These assumptions imply finite ancestry: following parents backwards from a fixed vertex cannot continue indefinitely.

An ancestor or descendant is always strict: an organism is not its own ancestor. Write $\mathrm{Desc}(v)$ for all descendants of v in G. For a nonempty subset $S \subseteq G$, the relevant conditions are as follows.

- **Connectedness (CONN):** the undirected graph induced on S is connected.
- **Identical ancestor point (IAP):** for each $v \in S$, at least one of $S \cap \mathrm{Desc}(v)$ and $S \setminus \mathrm{Desc}(v)$ is finite.
- **Convexity (CONV):** every vertex having both an ancestor and a descendant in S belongs to S.
- **Reflection (REF):** if $v \in S$ has infinitely many descendants in G, it has infinitely many descendants in S.
- **Common ancestor (CA):** some $r \in S$ is an ancestor of every member of $S \setminus \{r\}$.

IAP describes the internal distribution of descendants; REF compares the internal genealogy with the surrounding biosphere. They do different work. In particular, REF prevents a qualifying cluster from discarding almost all of the future of one of its members.

A **founder of** S is a member with no parent in S. This is relative to S: an organism can be a founder of a cluster while having parents outside it. Denote the founder set by $R(S)$. Every

member of S is a founder or descends, along a path inside S, from a founder. To see this, follow internal parents backwards; finite ancestry makes the process stop.

Define the **founding-cohort condition** by

$$\mathrm{FC}(S): \qquad t(r) < t(x) \quad \text{for every } r \in R(S),\ x \in S \setminus R(S).$$

If a connected S has more than one vertex, it has a non-founder. Under FC, all its founders precede this one fixed organism, so there are only finitely many founders. A connected set with no non-founder is a singleton, and also has finitely many founders.

The families we compare are

$$\mathcal{C} = \mathrm{IAP} \cap \mathrm{CONV} \cap \mathrm{CA} \cap \mathrm{REF},$$

$$\mathcal{F} = \mathrm{CONN} \cap \mathrm{IAP} \cap \mathrm{CONV} \cap \mathrm{REF} \cap \mathrm{FC},$$

and, for a fixed integer $N \geq 1$,

$$\mathcal{F}_N = \{S \in \mathrm{CONN} \cap \mathrm{IAP} \cap \mathrm{CONV} \cap \mathrm{REF} : |R(S)| \leq N\}.$$

Only nonempty sets are included. A maximal member means maximal by inclusion in the specified family.

Convexity and CA imply connectedness and FC. If r is the common ancestor, paths from r to other members remain inside S; r is the unique founder. Conversely, if a set has exactly one founder, the internal-parent argument makes that founder ancestral to all its other members. Consequently

$$\mathcal{C} = \mathcal{F}_1, \qquad \mathcal{C} \subseteq \mathcal{F}, \qquad \mathcal{C} \subseteq \mathcal{F}_N.$$

The connectedness and FC requirements in the first equality are automatic on the CA side. The families $\mathcal{F}$ and $\mathcal{F}_N$, $N \geq 2$, need not be nested.

3. Why finite founders suffice in the union argument

Alexander calls a family upward generic when it is closed under unions of nonempty ascending chains [A, Definition 7]. Such closure provides the upper bounds needed for Zorn's lemma. Here the difficult condition is IAP: a union can accumulate infinitely many descendants and non-descendants even when no single member does.

The following lemma isolates the part of Alexander's proof of Theorem 9(4) that is needed. His common ancestor supplies one starting vertex. Finitely many founders supply one by a pigeonhole argument.

Lemma 1. IAP at a chain union with finitely many founders

Let C be the union of a nonempty chain of connected sets satisfying IAP, CONV and REF. If C has finitely many founders, then C satisfies IAP.

Proof. Connectedness and convexity pass to C, since each path or pair of endpoints involved is contained in one sufficiently large chain member. Reflection passes as well: a chain member containing a given vertex already supplies its infinitely many descendants.

Suppose IAP fails at $v \in C$. Infinitely many members of C are descendants of v, and infinitely many are not. Every member of C descends internally from one of its finitely many founders, or

is a founder itself. Therefore some founder w_1 has infinitely many descendants in C that do not descend from v.

Because w_1 has finitely many children, one child w_2 has infinitely many of these descendants. Convexity puts w_2 in C. It is not a descendant of v, since otherwise all its descendants would be descendants of v. Repeating gives an infinite path

$$w_1 \longrightarrow w_2 \longrightarrow w_3 \longrightarrow \cdots$$

inside C, entirely outside the descendants of v.

Choose a chain member C_α containing v, w_1. The infinite path shows that w_1 has infinitely many descendants in G; REF makes C_α infinite. Fix $n > 1$, and choose a chain member $C_\beta \supseteq C_\alpha$ containing w_n. Reflection and the rest of the path give infinitely many descendants of w_n in C_β. IAP there makes its non-descendants finite. Thus the infinite subset C_α contains a descendant of w_n. It also contains the ancestor w_1, so convexity puts w_n in C_α.

This holds for every n. Consequently C_α contains infinitely many non-descendants of v. Reflection also gives it infinitely many descendants of v, since v has infinitely many in $C \subseteq G$. This contradicts IAP for C_α. □

The lemma assumes finiteness of the founder set of the **union**. Finiteness separately at each stage is insufficient: the number can increase without bound along the chain. Each of our two constraints supplies the missing union condition in its own way.

4. Two coverage theorems

Theorem 2. Founding cohorts

The family $\mathcal{F}$ is upward generic. Every organism belongs to a maximal member of $\mathcal{F}$.

Proof. Let C be a nonempty chain union from $\mathcal{F}$. Connectedness, convexity and reflection pass to C as in Lemma 1. To verify FC, take a founder r of C and a non-founder x with an internal parent p. Some chain member contains r, x, p. In that member r is still a founder and x is a non-founder, so $t(r) < t(x)$. Thus C satisfies FC and has finitely many founders by Section 2. Lemma 1 supplies IAP.

For coverage, Alexander's Theorem 13 supplies a member of $\mathcal{C}$ containing any specified organism. It belongs to $\mathcal{F}$. The subfamily of $\mathcal{F}$ containing that member has upper bounds for all nonempty chains; Zorn's lemma gives a maximal member. Any larger member of $\mathcal{F}$ would still contain the specified seed, so this is maximal in the whole family. □

Theorem 3. Fixed finite founder bounds

For every fixed $N \geq 1$, the family $\mathcal{F}_N$ is upward generic, and every organism belongs to a maximal member. At $N = 1$ this is Alexander's common-ancestor family.

Proof. Let C be a nonempty chain union from $\mathcal{F}_N$. If C had $N + 1$ founders, one chain member would contain them all. They would still be founders there, contradicting the bound. Hence $|R(C)| \leq N$. Connectedness, convexity and reflection pass to the union, and Lemma 1 gives IAP.

The same seed from Alexander's Theorem 13 lies in $\mathcal{C} \subseteq \mathcal{F}_N$. Zorn's lemma gives coverage. The equality $\mathcal{F}_1 = \mathcal{C}$ was proved in Section 2. □

These theorems answer the request after [A, Theorem 13] for an alternative without CA. They use his coverage theorem to obtain initial clusters and adapt his chain method to extend them. They do not claim a replacement proof of that published initial-existence theorem.

Examples and the role of the bound

The late-founder genealogy in Section 1 satisfies IAP: every vertex has all but finitely many vertices as descendants. The whole biosphere is connected, convex and reflective, has two founders, and is maximal because it is all of G. It is a member of every $\mathcal{F}_N$, $N \geq 2$, but not of $\mathcal{F}$.

A founding-cohort example can be completely explicit too. Take two parentless organisms a, b, both parents of c_0, followed by $c_0 \to c_1 \to \cdots$. Give a, b the first two birthdates. The whole graph satisfies FC, IAP, CONV and REF, and has no common ancestor. It is therefore a maximal member of $\mathcal{F}$. Its two CA-maximal subsets are $G \setminus \{a\}$ and $G \setminus \{b\}$. Starting at a later c_i gives smaller qualifying sets, each contained in both of those larger ones.

The distinction between a fixed bound and an unspecified finite number is witnessed by a different genealogy. Let

$$v_0 \to v_1 \to v_2 \to \cdots, \qquad w_i \to v_{i+1} \quad (i \geq 0),$$

with birthdates $t(v_i) = 2i$, $t(w_i) = 2i + 1$. For $m \geq 0$, put

$$S_m = \{v_i : i \geq 0\} \cup \{w_i : 0 \leq i < m\}.$$

Each S_m is connected, convex and reflective. It has $m + 1$ founders and satisfies IAP: a backbone vertex misses only finitely many earlier vertices and immigrants; every included immigrant also misses only finitely many vertices. But the union contains infinitely many descendants and non-descendants of v_0, namely the later backbone vertices and all the w_i. Its IAP fails.

In this entire graph, any reflective IAP set containing v_0 contains only finitely many immigrants: reflection forces infinitely many backbone descendants, and IAP then excludes infinitely many immigrants. It is therefore contained in some S_m and can be enlarged to a larger S_{m+1}. Thus the family of connected, convex, reflective IAP sets with finitely many founders, without one common bound, has no maximal member containing v_0. A fixed bound prevents this particular accumulation; FC prevents it by its temporal condition. The example places no prohibition on an individual late founder.

5. The infinite species classes are preserved

Different founding conventions can change the admitted finite histories. We now ask whether they change the infinite species classes in Alexander's sense.

For an infinite set S, define

$$\mathrm{gen}(S) = \{v \in S : S \setminus \mathrm{Desc}(v) \text{ is finite}\}.$$

These are its generators. Alexander defines $S \sim T$ when

$$\mathrm{gen}(S) \triangle \mathrm{gen}(T) \quad \text{is finite.}$$

The definition concerns **infinite** sets. It does not identify finite maximal clusters.

We use Alexander's Objective Species Theorem [A, Theorem 6]: two infinite specieslike clusters with infinitely many common generators are equivalent under $\sim$.

Theorem 4. Agreement of represented infinite classes

The infinite maximal members of $\mathcal{F}$, of each $\mathcal{F}_N$, and of $\mathcal{C}$ represent exactly the same $\sim$-classes.

Proof. Let S be an infinite member of either alternative. It has finitely many founders. Some founder r has infinitely many descendants in S, and IAP at r makes

$$S_r = \{r\} \cup (S \cap \mathrm{Desc}(r))$$

cofinite in S.

The set S_r has common ancestor r. IAP is inherited by subsets. Convexity follows from convexity of S and transitivity of ancestry. If $x \in S_r$ has infinitely many descendants in G, REF for S puts infinitely many in S, all descended from r, so REF holds in S_r. Thus $S_r \in \mathcal{C}$.

By [A, Theorem 9(4)] and Zorn's lemma, extend S_r to a maximal $M \in \mathcal{C}$. Finite branching and convexity give an infinite directed path in S_r. Every vertex on that path has infinitely many descendants in both S and M; IAP makes it a generator of both. Their common generator set is infinite. The Objective Species Theorem gives $S \sim M$.

Conversely, let M be an infinite maximal member of $\mathcal{C}$. It belongs to both alternatives, and Theorems 2 and 3 extend it to a maximal member S of either one. An infinite path in M again consists of common generators of M, S. The same published theorem gives $M \sim S$. □

The first direction works even when S is not maximal. Maximality specifies the collections of representatives being compared, rather than supplying that direction's argument. The theorem concerns finite symmetric difference of **generator sets**. It neither asserts finite symmetric difference of the clusters themselves nor makes maximal representatives unique.

6. What finite observations can tell us

The coverage theorems are existence statements about a whole infinite biosphere. Their use of an infinite future is substantial. A genealogical record can be extended tomorrow along a side lineage that has not yet permanently died out; no finite record settles whether that extension will happen infinitely often.

To make this observation precise, number vertices by birth order, $0, 1, 2, \ldots$, and set $t(n) = n$. Coordinate n specifies the subset of $\{0, \ldots, n-1\}$ that parents n. Each coordinate has finitely many possibilities. Give the product the topology in which an initial finite record is an observation.

Let $\mathcal{B}$ be the subspace with one root, vertex 0, exactly one parent for every other vertex, and at most two children per vertex. Violations have finite witnesses, so $\mathcal{B}$ is closed. Every member is a connected infinite biosphere.

A set is F_σ if it is a countable union of closed sets. A $\mathbf{\Pi}^0_3$ set is a countable intersection of F_σ sets. Completeness here means that every such subset of Cantor space continuously reduces to the property in question. A continuous reduction transforms records so that each finite output record depends on only finitely much input. The boldface notation denotes **Borel complexity**, not an arithmetical index set or a uniform algorithm for arbitrary presentations.

7. The complexity of global IAP

Theorem 5. Third-level Borel completeness

Global IAP is $\mathbf{\Pi}^0_3$-complete on $\mathcal{B}$. On the larger space of birth-coded biospheres it has the same relative upper bound and is at least this hard.

Proof: upper bound. Let $D(v, n)$ mean that v is a strict ancestor of n. This is determined by the first $n + 1$ coordinates. IAP is exactly

$$\forall v \; \exists (N, b) \in \mathbb{N} \times \{0, 1\} \; \forall n \geq N \; [D(v, n) = b].$$

For fixed v, N, b, the universal condition is closed. The existential choice gives an F_σ set, and the outer intersection gives $\mathbf{\Pi}^0_3$.

Proof: lower bound. Consider

$$P_3 = \{x \in 2^{\mathbb{N}\times\mathbb{N}} : \text{each row of } x \text{ has finitely many ones}\}.$$

This is a complete $\mathbf{\Pi}^0_3$ set; a proof is supplied in Appendix A. We turn its rows into side chains attached successively to a permanent backbone.

At stage s, create b_s, with parent b_{s-1} if $s > 0$; create a_s with parent b_s; then, for $k = 0, \ldots, s$, inspect $x(k, s - k)$. When it is one, append a vertex to the current tip of the side chain starting at a_k. A zero appends nothing. Give all created vertices consecutive birth numbers.

The graph G_x has one root, every other vertex has one older parent, and each vertex has at most two children. Its permanent backbone makes it infinite. Each finite output record depends on finitely many input bits, so $x \mapsto G_x$ is continuous.

If row k has infinitely many ones, a_k has infinitely many descendants in its side chain and infinitely many non-descendants on the backbone. Global IAP fails.

If every row has finitely many ones, each side chain is finite. A vertex on a side chain therefore has finitely many descendants. For b_k, the non-descendants lie in the finitely many earlier backbone vertices and the finitely many finite side chains with index below k, together with b_k itself. Thus its non-descendants are finite. Every vertex satisfies IAP, proving

$$G_x \in \text{IAP} \quad \Longleftrightarrow \quad x \in P_3.$$

This gives the required continuous reduction. □

The construction uses a single founder throughout. Allowing several founders is therefore not the source of this complexity. On a locally finite rooted tree the same issue can be viewed through infinite rays: two rays that separate permanently provide infinitely many descendants and non-descendants; a unique infinite ray makes each off-ray subtree finite. This connects the calculation with the familiar distinction between one infinite end and several.

Corollary 6. No uniform eventual yes/no learner

There is no finite-record learner that eventually gives the correct yes/no answer for IAP at a designated organism on every member of $\mathcal{B}$. The learner need not be computable.

Proof. Set all rows except row 0 to zero in the preceding construction. IAP at a_0 holds exactly when that row has finitely many ones. Let FIN be this subset of Cantor space. It is countable and

dense and is not G_δ: were it a dense countable intersection of open sets, it would be comeager as well as meager, contradicting the Baire category theorem.

If a yes/no learner's answers g_s, based on finite records, converged to the right answer on every input, then

$$\mathrm{FIN} = \bigcup_N \bigcap_{s \geq N} \{g_s = 1\}$$

would be F_σ. The same formula with zero would make its complement F_σ, so FIN would also be G_δ. This is impossible. The same restricted reduction in Theorem 5 also rules out a learner deciding global IAP on these records. □

This conclusion is about a specified binary decision problem. A species-valued output has different success criteria; no impossibility theorem for all such outputs follows without defining those criteria.

8. Consequences and scope

The two founder constraints change which founding histories can occur in a maximal cluster while preserving Alexander's represented infinite species classes. The founding-cohort condition gives a temporal convention; the bound N gives a numerical convention and permits late founders. Neither theorem establishes which convention biologists should choose.

Theorem 4 is a consequence of Alexander's Objective Species Theorem and the finite-founder constructions. Theorem 5 and Corollary 6 concern the observation space explicitly defined here. No claim of uniform computability from arbitrary infinite graph descriptions, or of a practical classification algorithm, is made.

All arguments are hand proofs. Alexander's published theorems are credited where used. The explicit complexity reduction is included for this formulation; its relation to known one-ended-tree phenomena is relevant when assessing historical novelty. Independent human review and priority have not been established.

Appendix A. Why the row-finiteness set is complete

Let $A \subseteq 2^{\mathbb{N}}$ be any $\mathbf{\Pi}^0_3$ set. Write

$$A = \bigcap_k \bigcup_N F_{k,N}, \qquad F_{k,N} \text{ closed}, \quad F_{k,N} \subseteq F_{k,N+1}.$$

For input z and stage s, let $m_k(s)$ be the least $N \leq s$ such that the cylinder determined by $z \upharpoonright s$ meets $F_{k,N}$, and put $m_k(s) = s + 1$ if there is none.

For fixed k, this integer sequence is nondecreasing. Excluded candidates stay excluded as the cylinder shrinks, and a newly available candidate has index larger than every previously available one. If $z \in F_{k,N}$, that candidate bounds the sequence once $s \geq N$. If z belongs to none of the $F_{k,N}$, closedness eventually excludes every fixed candidate, so the sequence is unbounded.

Output one in row k exactly when $m_k(s)$ increases, starting with $m_k(-1) = 0$. Each output bit depends on a finite prefix of z. Its row has finitely many ones exactly when the integer sequence is bounded. Therefore this is a continuous reduction of A to P_3. The direct zero-tail definition of row-finiteness gives the upper bound. No effective presentation of the closed sets is assumed in this topological argument. □

References

[**A**] Samuel Allen Alexander. *Specieslike clusters based on identical ancestor points.* Journal of Mathematical Biology (2026). DOI: 10.1007/s00285-026-02361-x. Source version: arXiv:2602.05274v1, 5 February 2026. Definitions 1–6 and 8–10; Theorems 6, 9 and 13; the request following Theorem 13.